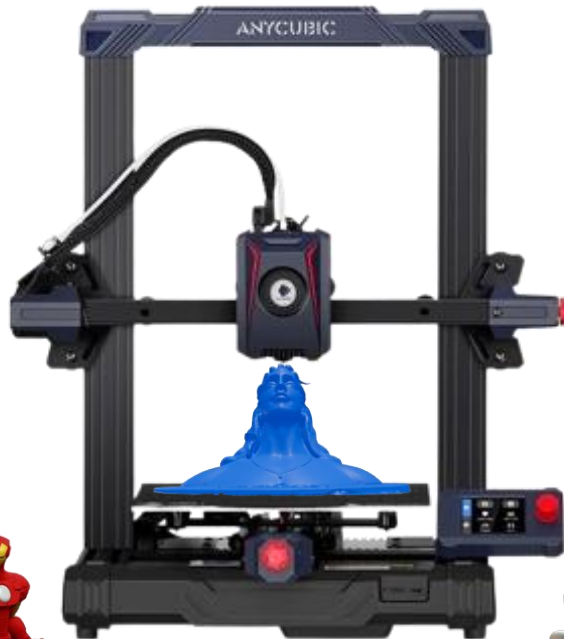


3D PRINTING COURSE



**TEACHER:
CERTIFIED
TRAINER**

**DURATION:
16 CLASSES
(1 HOUR PER CLASS)**

**MODE:
OFFLINE**

3D printing is the process of creating real, three-dimensional objects from a digital design. Unlike regular paper printing, which works in two dimensions, 3D printing adds depth, allowing us to produce physical models with volume and structure. From phone cases and toys to industrial tools, this technology is revolutionizing how we create and build. It is widely used in industries such as healthcare, construction, aerospace, food, and engineering.

At our centre, students typically complete 8–10 practical models, based on their individual pace and understanding. These models help them grasp the core concepts of 3D printing technology and give them real-time experience in building, designing, and innovating using modern tools.

Level 1: Introduction to 3D Printing & Basic Design (Ages 8–12)

1. What is 3D Printing?

- Basic explanation of 3D printing
- Observe a cube being printed live
- Understanding the concept of layers and how objects are formed

2. How a 3D Printer Works

- Introduction to parts: Filament, hotend, bed
- Bed leveling basics
- Filament loading and preheating

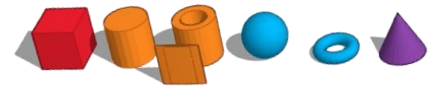
3. Introduction to 3D Design (Tinkercad)

- Navigating Tinkercad software
- Using tools like move, rotate, and scale
- Understanding grid and placement



4. Creating Basic Shapes

- Making cubes, cylinders, spheres
- Hands-on: Design a basic object using Tinkercad

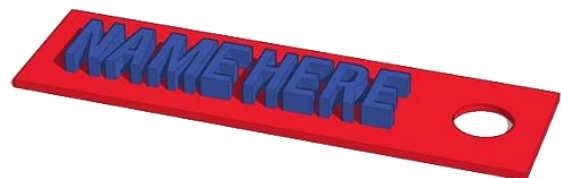


5. Combining Shapes to Create Objects

- Learning to group and merge
- Hands-on: Personalized name keychain

6. Designing an Open Box

- Learn to create hollow objects
- Hands-on: Print a small open box



7. Slicing Software (Cura)

- Introduction to Cura software
- Exporting STL files and slicing for printing

8. Understanding Print Settings

- Layer height, infill percentage, support structures
- How settings affect time and quality

9. Creating a Functional Object

- Design for usability
- Hands-on: Design and print a phone stand

10. Supports & Overhangs

- Learn when supports are needed
- Print a model with overhangs

11. Mini Puzzle Game Design

- Understanding tolerance
- Design & print a simple maze puzzle

12. Interlocking Parts

- Design fit and spacing for connection
- Print Lego-style bricks

13. Organizing Your Desk

- Functional design for daily use
- Hands-on: Stackable desk organizer

14. Basic Troubleshooting

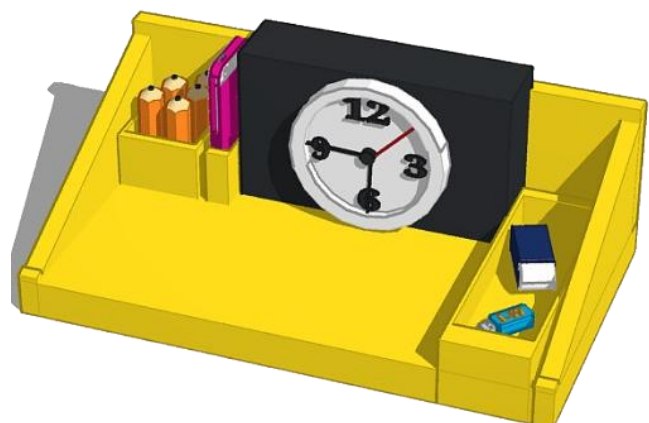
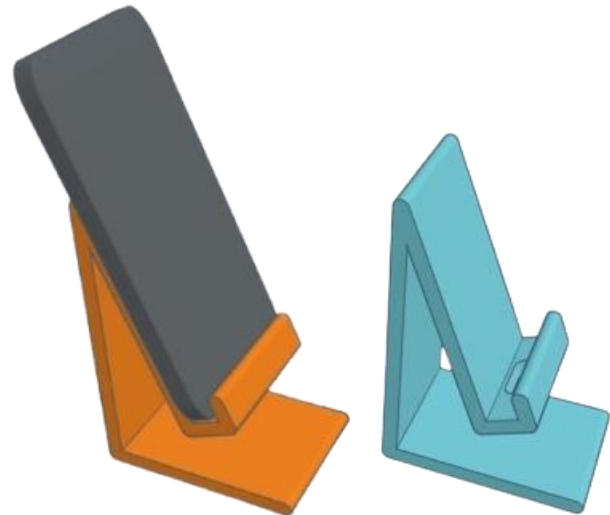
- Common print issues and their causes
- How to fix and avoid print failures

15. Final Project (Student's Choice)

- Design a custom model from scratch
- Prepare it for print

16. Show & Tell Presentation

- Explain design, print process, and learning
- Present final model to peers



Level 2: Fusion 360 + Blender + Advanced Printing (Ages 12–16)

1. Advanced 3D Printing Techniques

- Understanding multi-part models
- Observe demo prints with moving parts

2. Introduction to Fusion 360

- Interface, navigation, sketch basics
- Hands-on: Print a phone stand



Fusion 360

3. Parametric Design & Constraints

- Using dimensions and rules
- Hands-on: Snap-fit box design

4. Loft & Fillet Tools for Complex Shapes

- Curves, smooth edges, and transitions
- Design a rounded case



5. Precision in Design

- Using calipers for measurement
- Design a custom iPhone case

6. Getting Started with Blender

- Navigation and UI overview
- Create a simple shape

7. Text and Engraving in Blender

- Adding 3D text
- Modify the keychain with engraved name

8. Modifying STL Files

- Importing and editing existing files
- Add features to downloaded models



9. Engineering Design Concepts

- Design components that fit together
- Print a multi-part pen holder

10. Tolerance Testing

- Measure and refine fit
- Test fit of printed parts

11. Functional Home Tools

- Think and design useful items
- Hands-on: Custom household tool

12. Fusion 360 Joints & Assemblies

- Working with multiple moving parts
- Design a snap-fit enclosure

13. Preparing Large Models

- Splitting for batch printing
- Combine post-print

14. Final Project Planning

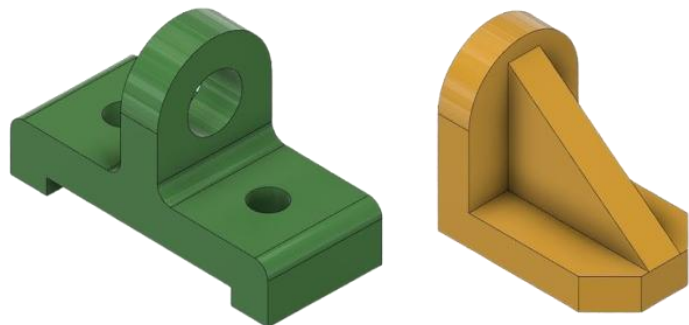
- Brainstorm idea
- Start 3D design prototype

15. Final Model & Adjustments

- Troubleshoot and refine prints
- Finalize model

16. Project Presentation

- Showcase completed project
- Explain design journey and applications



Level 3: Advanced Projects & Real-World Applications (Ages 16–18)

1. Mechanical Design Concepts

- Learn gear systems and motion
- Print a working gear mechanism



2. Designing for Strength

- Understand wall thickness and infill
- Print a strong custom key holder

3. Modular Design Techniques

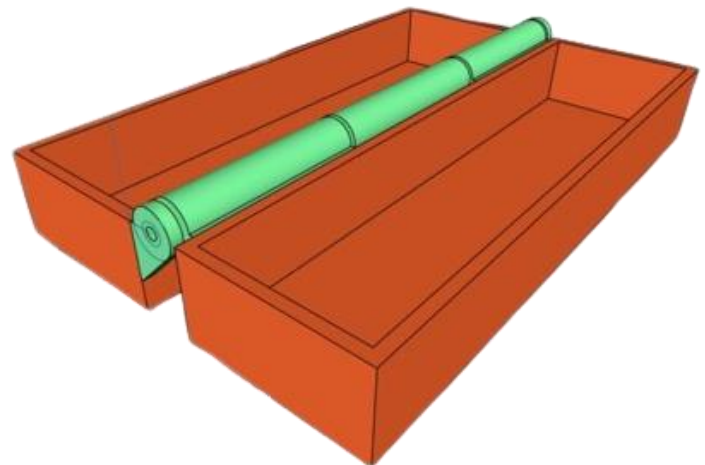
- Build systems with attachable modules
- Hands-on: Stackable storage box

4. Snap-Fit & Hinged Models

- Create enclosures that open/close
- Print a hinged electronics case

5. Multi-Part Assembly

- Design parts that function together
- Build and print a hinged box



6. Architectural Modeling

- Learn scale and layout
- Students design buildings for a mini city

7. Threaded Parts & Screws

- Print functional screw cap containers
- Learn thread accuracy

8. Everyday Use Jigs & Fixtures

- Print tools and stands

- Hands-on: Phone or pen stand

9. Medical & Engineering Use Cases

- Design basic prosthetics or bio-tools
- Print a simple prosthetic grip

10. Robotics & Automation

- Design for robotic components
- Print: Robotic arm joint

11. Gears & Motion Transfer

- Print and test gear systems
- Understand torque and fit

12. Electronic Enclosure Design

- Create Arduino or Raspberry Pi cases
- Focus on wiring paths and ventilation

13. Final Project Planning

- Ideate and draft personal project
- Begin 3D modeling

14. Prototyping & Iterations

- Test print, improve design
- Apply rapid prototyping techniques

15. Print Troubleshooting & Fixes

- Identify common errors
- Correct settings for better output

16. Showcase & Feedback

- Present complete project
- Receive peer and mentor review

